Scientific Justification .

Supermassive black hole binaries (SMBHBs) are thought to be natural consequences under the prevailing cosmological paradigm wherein massive galaxies 1) contain a supermassive black hole (SMBH) at their centers and 2) frequently undergo mergers. At milliparsec separations, these binaries become loud gravitational wave (GW) sources in the Pulsar Timing Array (PTA) band and likely produce a significant electromagnetic signal prior to merger. Such a system would be practically impossible to distinguish from an Active Galactic Nucleus (AGN) given spatial resolution limitations. However, in the low mass ratio, circular binary regime, orbital motion would sinusoidally modulate the rest frame brightness of the secondary SMBH's accretion disk via relativistic Doppler boosting. As such, detecting a sinusoidal signal in an AGN light curve would serve as evidence in support of a binary hypothesis.

Previous periodic AGN searches have been limited by the stochastic variability intrinsic to AGN and the cadence and duration of available light curve data. The inconsistent and sparse cadence of many modern survey strategies make detecting short-timescale periodicity in AGN light curves especially difficult. Witt et al. [2022] presents demonstrates the ability of the `ultranest` nested sampler to accurately disentangle a sinusoid from stochastically variable data. However, they find short-period periodicity ($P < 100$ d) can be subsumed into the stochastic variability fit as the sampler may equate the characteristic timescale of the stochastic process to the injected period [Witt et al., 2022] of the sinusoid. Recently, Banaszak et al. [2026] expands this method to a joint model selection scheme using the Generalized Lomb Scargle Periodograms, and investigates how this joint scheme performs for simulated LSST and Zwicky Transient Facility (ZTF) light curves. Particularly, Banaszak et al. [2026] finds the nested sampler correctly detects periodicity in $\sim$74% of simulated short-period ($P < 6$ months) SMBHB LSST light curves and falsely detects periodicity in 0.83% of SMBH LSST light curves. Turning to ZTF light curves, whose average r-band sampling cadence of $\sim$3.5 days is a vast improvement over LSST's $\sim$15 day cadence, this scheme detects $\sim$80% of short-period SMBHBs and returns a false positive rate of only 0.07%. This indicates the ZTF surveying strategy enables us to better discern short-timescale periodicity from stochastic fluctuations.

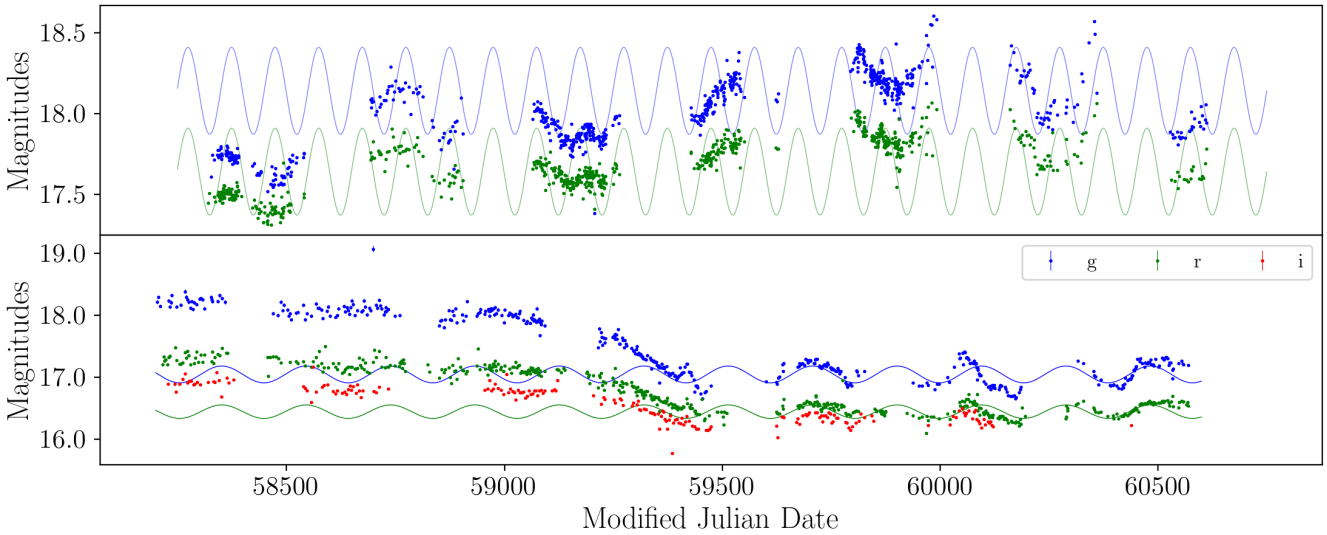

Figure 1: Two significant ZTF periodic AGN candidates with clear short-timescale variability

Furthermore, multi-wavelength AGN observations reveal intrinsic AGN variability in longer wavelength bands (ri) manifests as a delayed, downscaled realization of the variability in short wavelength optical bands (ug) [Kovačević et al., 2022]. Yu et al. [2026] presents `EzTaoX`, a sampler which can recover these inter-band lags and scaling factors. By inverting these transformations, this software allows us to combine multi-wavelength AGN light curves and effectively sample the underlying variability at a far more rapid cadence. Yu et al. [2026] demonstrates this procedure enables vastly improved recovery of stochastic variability parameters for particularly short timescales <50 d. Since the periodic signal from Doppler boosting shares a period and phase between bands, this procedure allows us to jointly constrain a shared periodic signal which is effectively exposed to phase-shifted and scaled variations of the intrinsic AGN variability. With more accurate characterization the stochastic fluctuations in a given light curve, short-timescale periodicities will become more apparent using this multi-wavelength period search.

Although the false positive rate found by Banaszak et al. [2026] was extremely low for ZTF light curves, the high expected population imbalance between SMBHBs and SMBHs could cause such a scheme to produce a relatively impure (precision $\sim$50%) sample of periodic AGN. Here, we seek to bolster the aforementioned multiband procedure laid out Yu et al. [2026] and Banaszak et al. [2026] by compounding ZTF's rapid g- and r- band observations with daily-cadenced i-band observations from Las Cumbres Observatory's (LCO) 1-meter telescope at the McDonald Observatory. The LCO is well-conditioned for targets of ZTF-level brightness in the ZTF sky area, so its use will provide the best opportunity to approve or reject the binary hypothesis on several periodic AGN candidates identified in the ZTF catalog. Specifically, we expect that this procedure, which effectively triples our observation cadence, will further bolster our joint model selection scheme beyond the performance achieved for the single band ZTF cadence.

Experimental Design *Describe your overall observational program. How will these observations contribute toward the accomplishment of the goals outlined in the science justification?*

We follow the road map provided in Banaszak et al. [2026] to analyze ZTF r-band light curves and identify ∼100 strong periodic AGN candidates for follow-up. We plan to use LCO's 1-meter telescope at the McDonald Observatory in Texas to take follow-up i-band observations on a subset of these candidates. Though we plan to merge these observations with archival ZTF data, we will not rely on ZTF to consistently image our targets in the i- band, as their cadence is unreliable. As such, we propose a daily i-band cadence for the remainder of ZTF3, which is scheduled for the upcoming semester thorugh December 31, 2026. This would require 258 total observations per candidate.

First, we trim our candidate pool to those light curves with average brightness < 18 mag, to avoid probing near ZTF's i-band 5σ depth of ∼20.4 magnitudes. Second, we select light curves whose recovered period *or* characteristic stochastic timescale is less than 6 months (to accommodate cases where periodicity was captured by the stochastic variability fit). This leaves us with a final pool of 4 viable candidates which have an average i-band magnitude of 17.1. Two are dispalyed in **??**. Below, we formulate our final time request given the exposure time required for our science goal.

We use the LCO exposure time calculator to compute exposure time required for a standard 5σ photometric observation under various conditions.[1] As shown in Table 1, a 5σ observation of one of these candidates requires between $1-4$ seconds of exposure time, bringing the cost of our science goal up to 258-774 seconds of observation time per candidate. To be conservative, we adopt a cost of 700 seconds per candidate. With this, **we propose our final request of 2800 seconds**, or ∼46 minutes of telescope time to achieve our science goal of evaluating the binary hypothesis for 4 periodic AGN candidates by supplementing ZTF observations to achieve a very rapid effective multi-wavelength light curve cadence.

Table 1: Exposure Time for Different Conditions

Conditions	Airmass	Moon Phase	Magnitude	Signal-to-Noise	Exposure Time (sec)
Pessimistic	1	New	18	5	4
Average	1.5	Half	17.6	5	2
Optimistic	2	Full	17.	5	1

[1] Las Cumbres Observatory Exposure Time Calculator

References

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